

MAN00035C



**School for Business
and Society**

Quantitative Analysis in Business Studies

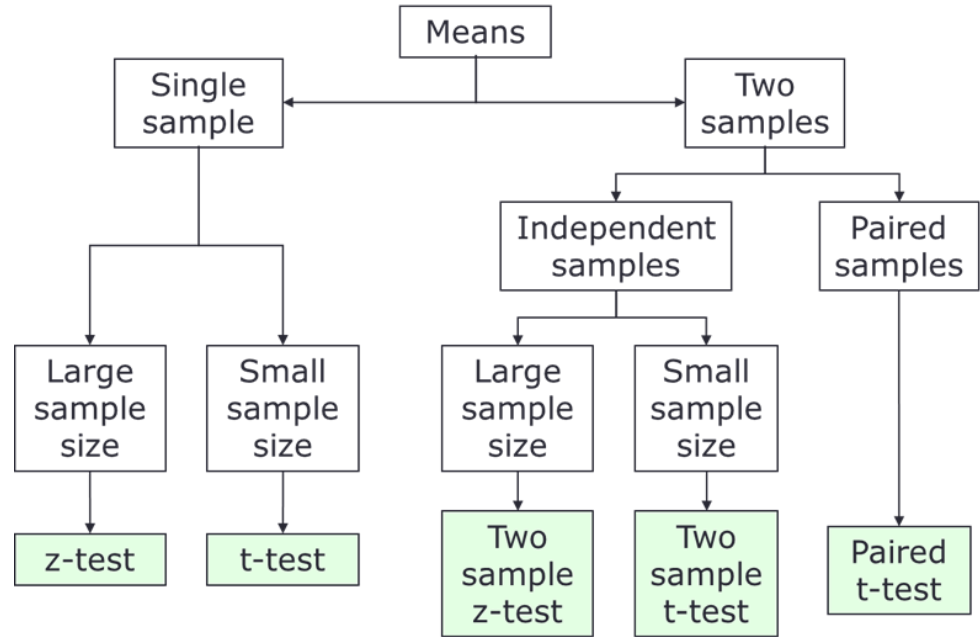
2026 - Week 7

Week 7: ANOVA

Reading: Taylor – Chapter 8

Where are we?

- Test for a mean/proportion, or the difference of two means (see the figure)
- Test if there is any association between categories in a two-way table. (Chi² test)
- This week: test the mean difference of more than 2 groups (Anova)



Steps for carrying out a hypothesis test

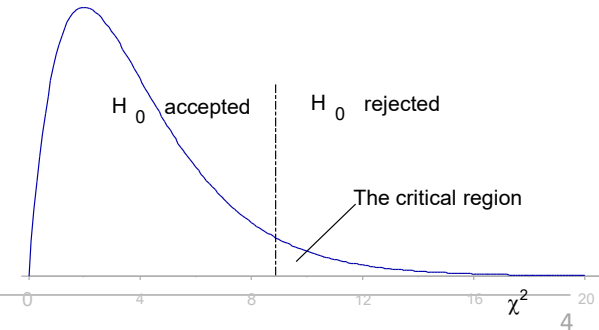
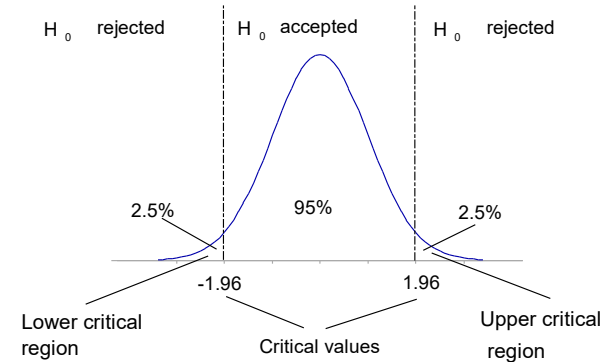
Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally $\alpha=5\%$) and determine the critical values

Step 3. Calculate the test statistic

Step 4. Decide whether to accept or reject the null hypothesis

- You need to discuss the assumptions and draw a conclusion as well



T-test vs. ANOVA (1)

- **2-sample t-test**

Means of **2 unrelated samples**:

e.g. Is there any difference between the mean starting salaries of men and women in a particular firm?

$$H_0: \mu_m = \mu_w$$

$$H_1: \mu_m \neq \mu_w$$

- **(One-way) Analysis of Variance**

Means of **more than 2 unrelated samples**:

e.g. Is there any difference between the average starting salaries for employees in four different departments in the firm?

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1: \mu_1 \neq \mu_2 \text{ or } \mu_1 \neq \mu_3 \text{ or } \mu_1 \neq \mu_4 \\ \text{or } \mu_2 \neq \mu_3 \text{ or } \mu_3 \neq \mu_4$$

T-test vs. ANOVA (2)

ANOVA can be thought of as an extension of the two-sample t-test to 3 or more groups of data.

(You can use it for just two groups too. It ends up the same as our “small sample” test, which is the SPSS “equal variances” one.)

Assumptions for T-test and ANOVA test:

1. observations are **independent**
2. **the variances are equal across groups** (homoscedasticity)
3. The data within each group should follow a **normal distribution**.

In t-test, we commonly considered a difference divided by a standard error in the last topic. That will change today. **(F-test)**

What would data look like?

Lets look at how suitable data might be represented in a dataset:

Height	College
1.56	A
1.78	V
1.54	C
1.70	D
1.70	J
1.83	H
1.73	G
1.63	L

So, we are **NOT** actually looking at two or more different continuous variables, but one, with the groups defined by another, categorical, variable.

The Layout

ANOVA follows the same thought process as we used last week

- Step 1. Set up the null ((no relationship, i.e $\mu_A = \mu_C = \mu_D$) and alternative hypotheses
 - **If we reject the null, we find that at least one group is different to the rest.**
 - **However, we don't know if it's one group or more than one, which group(s) they are, or ...**
- Step 2. Decide on the significance level (normally $p < 5\%$) and determine the critical values
- Step 3. Calculate the test statistic
- Step 4. Decide whether to accept or reject the null hypothesis

Why can't we just do a lot of two-sample t-tests?

- One test (5%): $P(\text{correct}) = 0.95$; $P(\text{incorrect}) = 0.05$
- More than one test for comparison:

No. of samples	Pairwise tests	$P(\text{all correct})$	$P\{\text{at least one incorrect}\}$
2	1	0.95	0.05
3	3	$0.95^3 = 0.875$	0.125
n	$\{n(n-1)\}/2$	0.95^n	$1 - 0.95^n$
10	45	0.95^{45}	0.90

- Obviously, this probability of error not acceptable.
- If the pairwise tests are not independent, we cannot even do this calculation.
- ANOVA compares all means simultaneously

Recap

Analysis of Variance

Standard deviation:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$$

Variance:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n-1}$$

Example 1

Find Sum of Squares $SS = \sum(x - \bar{x})^2$ for the following sample

A: {1, 3, 4, 4}

First calculate the mean!

$$\bar{x} = (1+3+4+4)/4 = 3$$

Then look at the sum of squares

$$= \sum(x - \bar{x})^2 = (1-3)^2 + (3-3)^2 + (4-3)^2 + (4-3)^2 = 6$$

Recap

Analysis of Variance

Standard deviation:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$$

Variance:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n-1}$$

Example 1

Find Sum of Squares $SS = \sum(x - \bar{x})^2$ for the following sample A: {1, 3, 4, 4}

$$SS=6$$

Example 2

Find Sum of Squares for the following sample A: {20, 10, 5, 5}

$$SS=150$$

Intuition to ANOVA (1)

ANOVA looks at some **measure of variation** of the data taking all the data points together, and splits that up into two types.

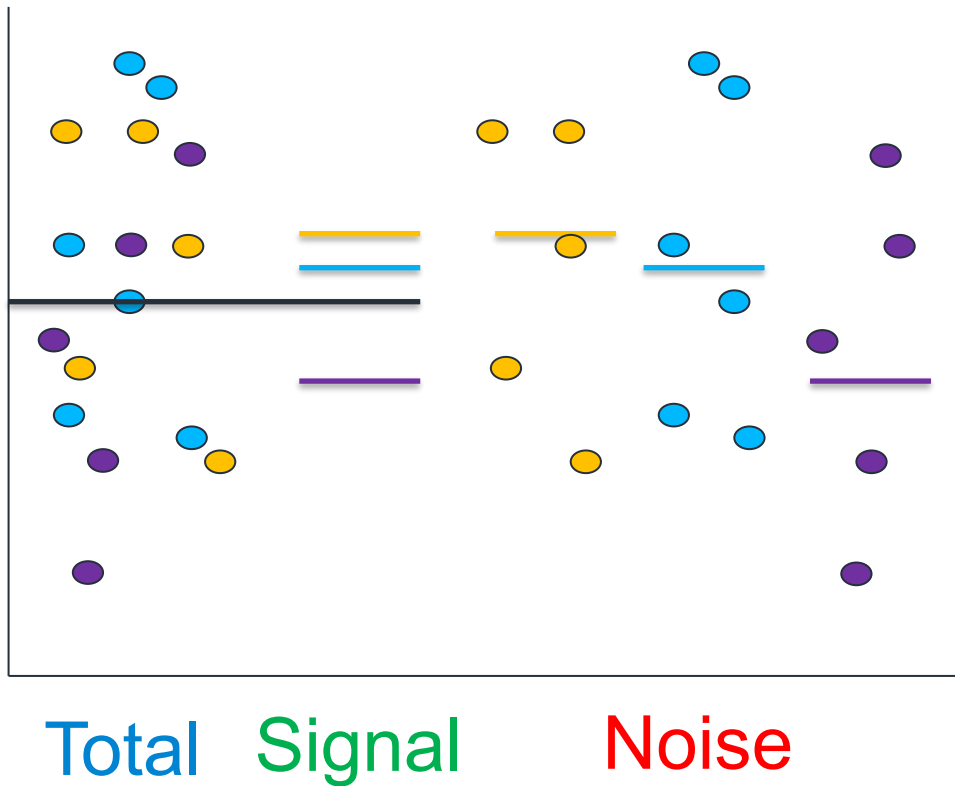
The chosen measure of variation is the “**sum of the squares of the differences**”.

The two types:

a) Variation **between** the group means (*a.k.a. actual differences between the groups, a **signal***)

b) Other variation, **within** the groups, put down to “errors” at this stage (*a.k.a. **noise***)

Intuition to ANOVA (2)

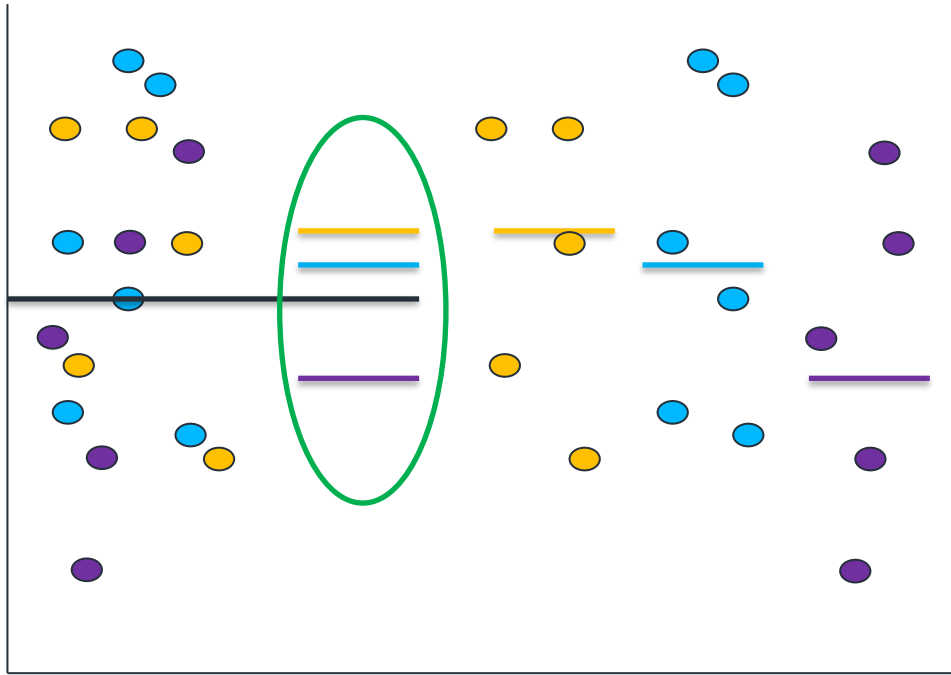


Some of the variation (the **signal**) in the data is due to the fact that the means truly are different.

Some of it is due to **noise**.

We are going to try to separate the two.

Intuition to ANOVA (3)

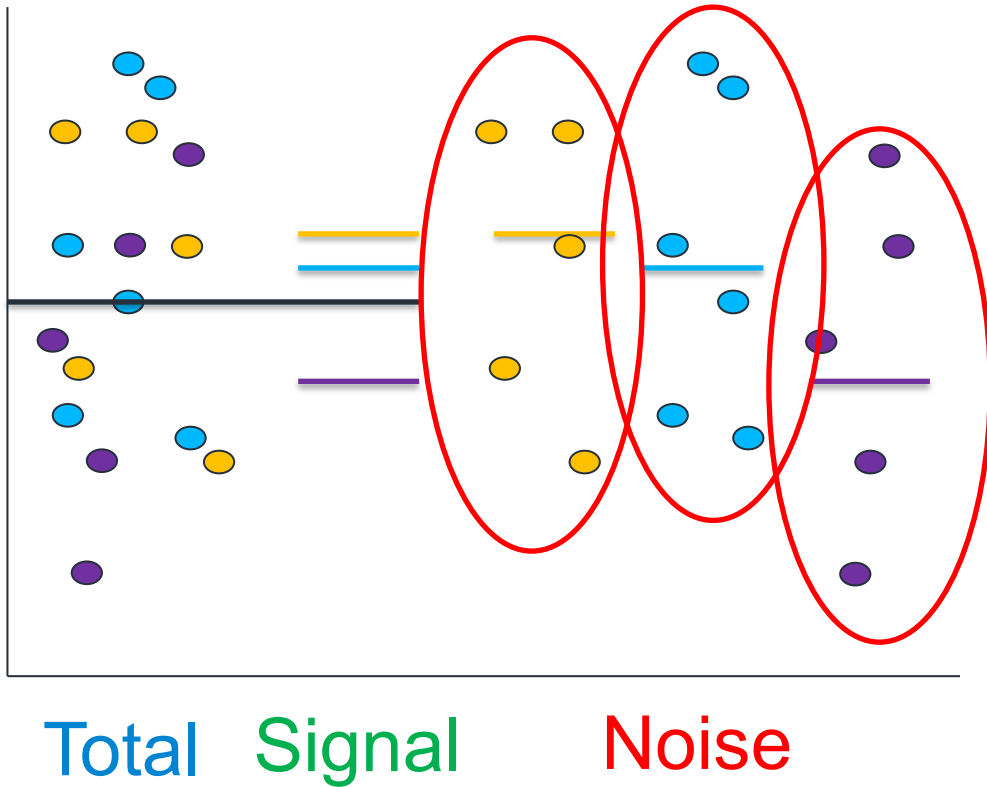


Total Signal Noise

The more different the means, the bigger the **signal** and the easier it is to see that they are actually different.

This is good for us.

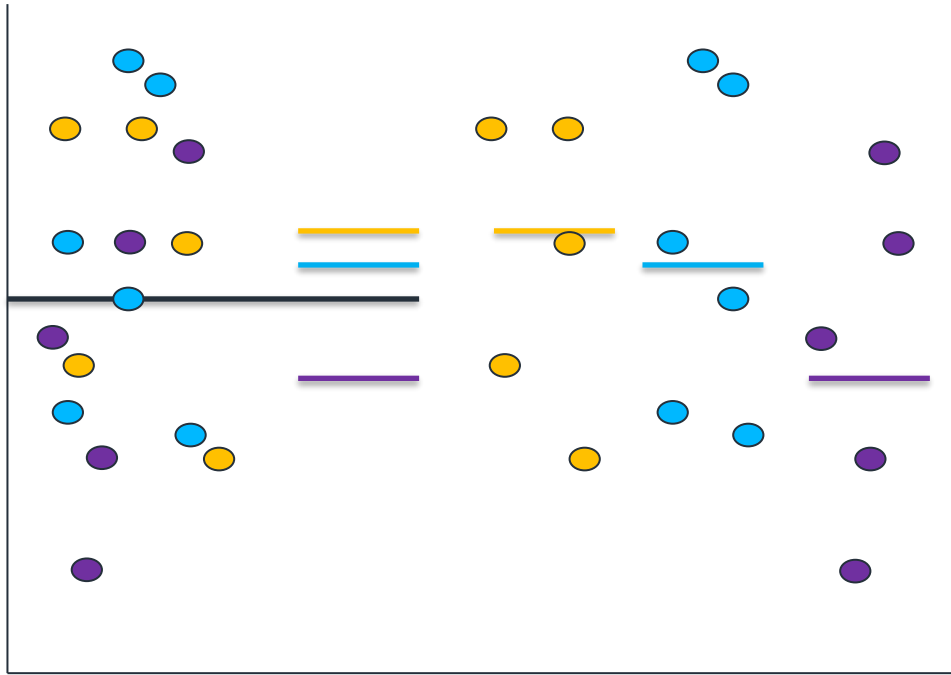
Intuition to ANOVA (4)



The more spread out the groups, the bigger the **noise** and the harder it is to be certain where the means should be (*think what happens to t -statistics when s increases*).

This is not helpful for us.

Two things to note (1)



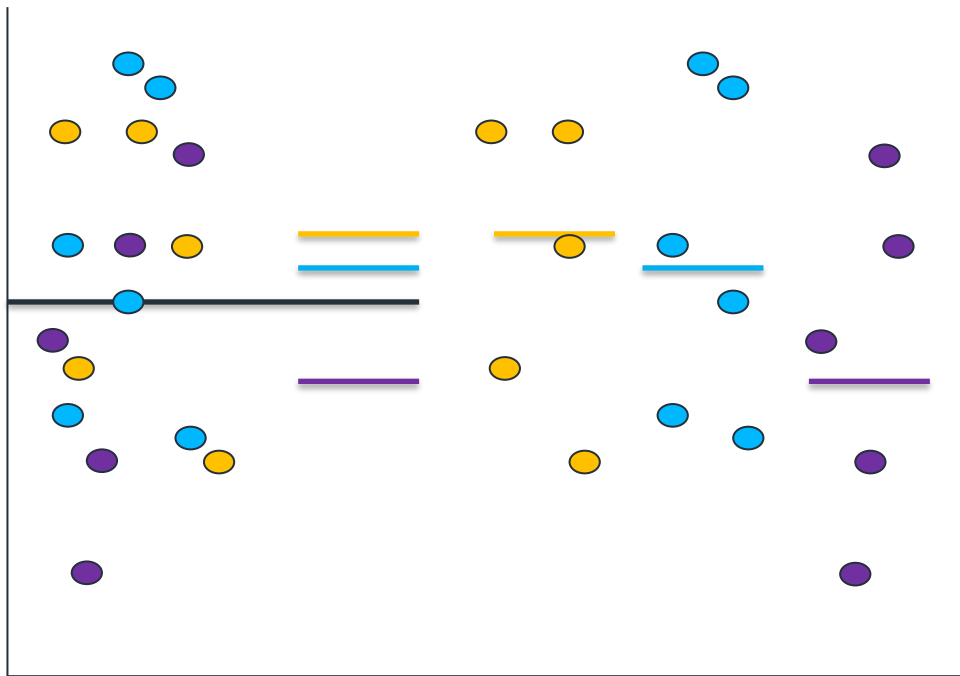
Total Signal Noise

However, it's worth noting a couple of things first.

1) We can change the **signal** without changing the **noise** by moving entire groups:

(So long as one counterbalances the other, this leaves the black line in the same place.)

Two things to note (2)



Total Signal Noise

However, it's worth noting a couple of things first.

2) We can change the **noise** without changing the **signal** by moving datapoints:

(So long as one counterbalances the other, this leaves the black line in the same place.)

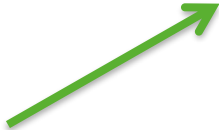
An Example (1)

It is easier to see what actually goes on with a suitable example.

There are two ways we can present the data.

This is how the computer would like to see it:

Scores from formative tests



Data	Group
2	A
4	A
4	B
5	B
6	B
2	C
3	C
9	C
10	C

This is probably better for us:



Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

An Example (2)

Our null hypothesis (*our assumption*) is that each group ***should*** have the same mean.

That is:

$$H_0: \mu_A = \mu_B = \mu_C$$

Our alternate is that this is not the case.

Note, the alternate is NOT

$$H_1: \mu_A \neq \mu_B \neq \mu_C.$$

We might have

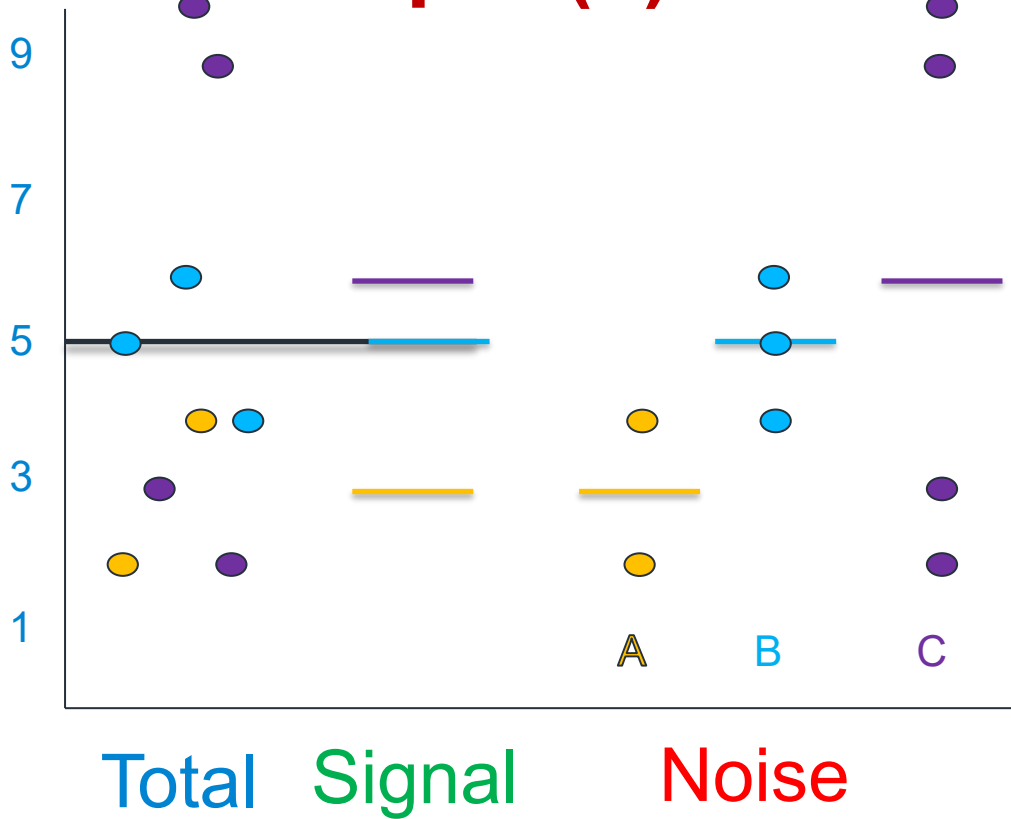
$$\mu_A = \mu_B \neq \mu_C, \text{ or}$$

$$\mu_A \neq \mu_B = \mu_C, \text{ or}$$

$$\mu_A = \mu_C \neq \mu_B.$$

All we would know is that they are not all the same.

An Example (3)

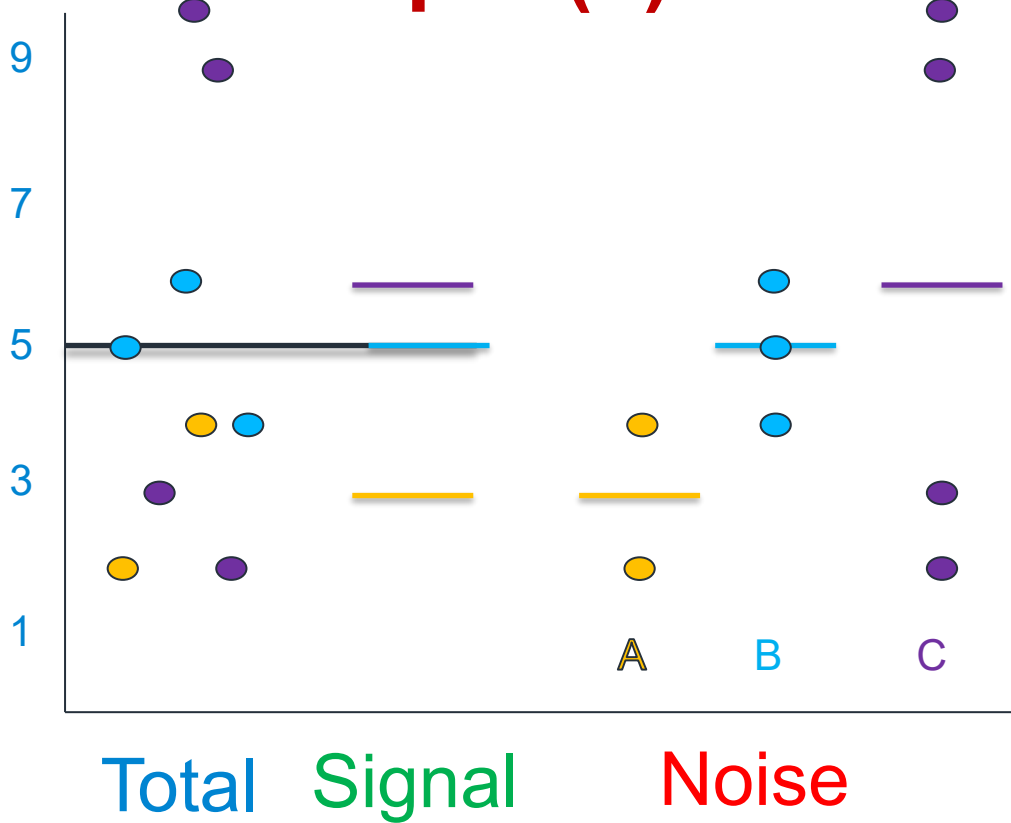


We can put this on a diagram.

(Not necessary for answering the question, but maybe helpful to illustrate what we are doing)

(We have also implicitly already calculated the mean of each group (3, 5 and 6 respectively).)

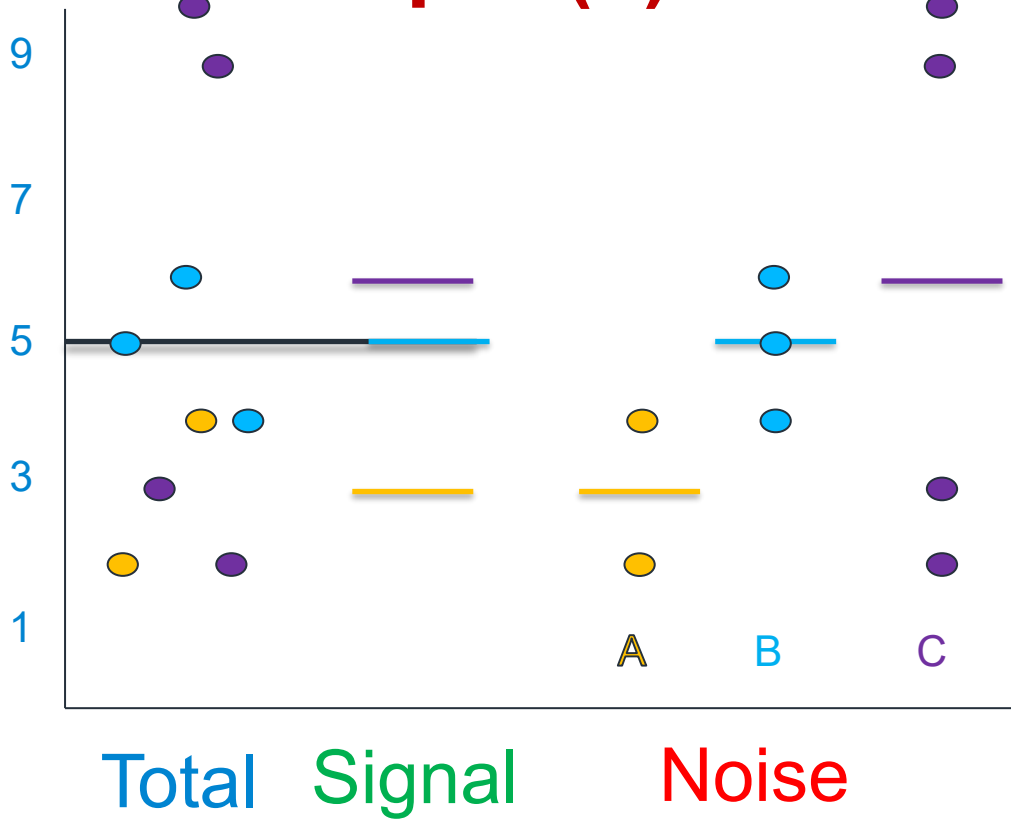
An Example (4)



Note that the mean of the datapoints and the (weighted) mean of the means of the groups are the same (the black bar).

This is a good check that you've found the group means correctly.

An Example (5)



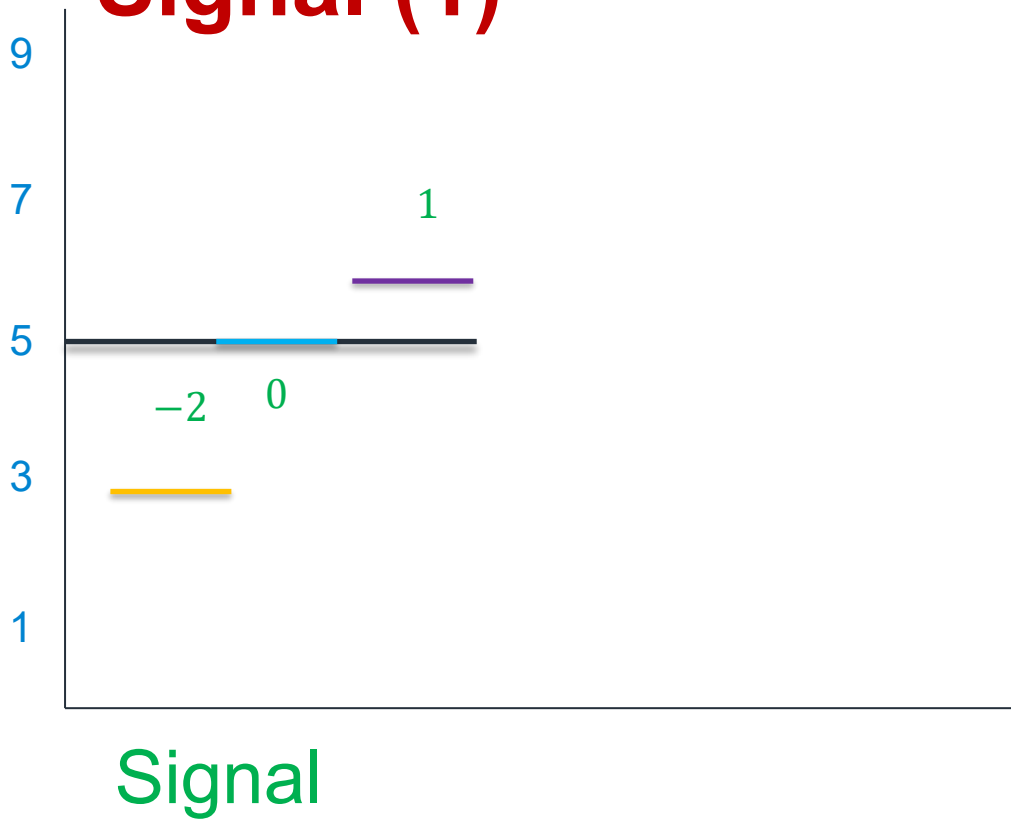
For each datapoint, we are going to look at three differences:

1) Group mean to grand mean

2) Datapoint to group mean

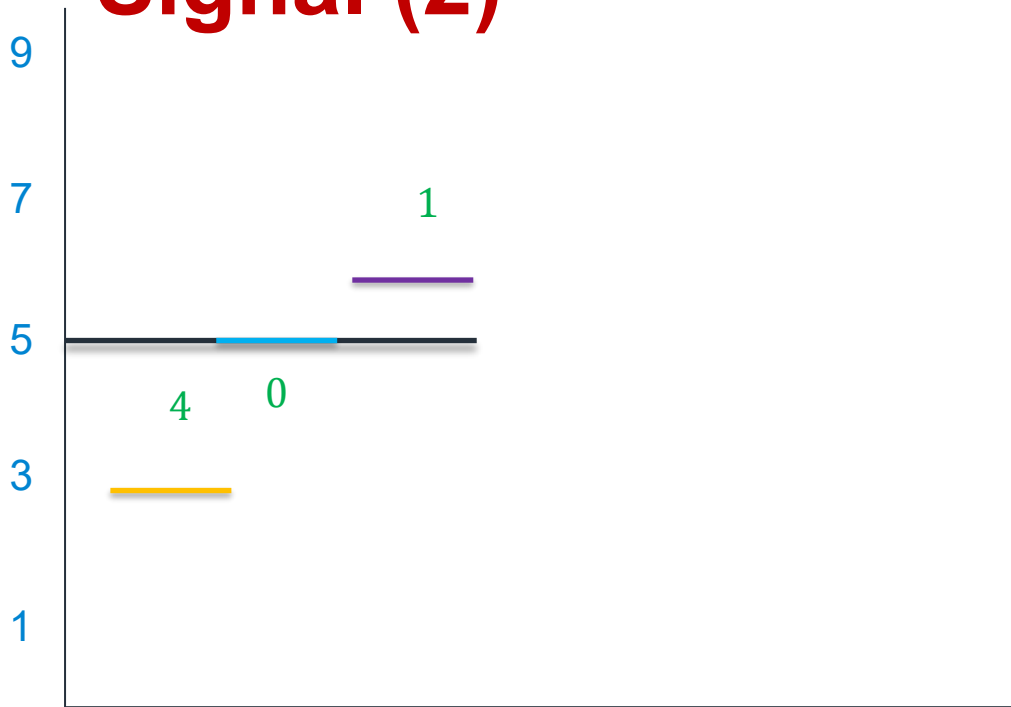
3) Datapoint to grand mean

Signal (1)



We look at each **distance** between **group means** and **the grand mean**

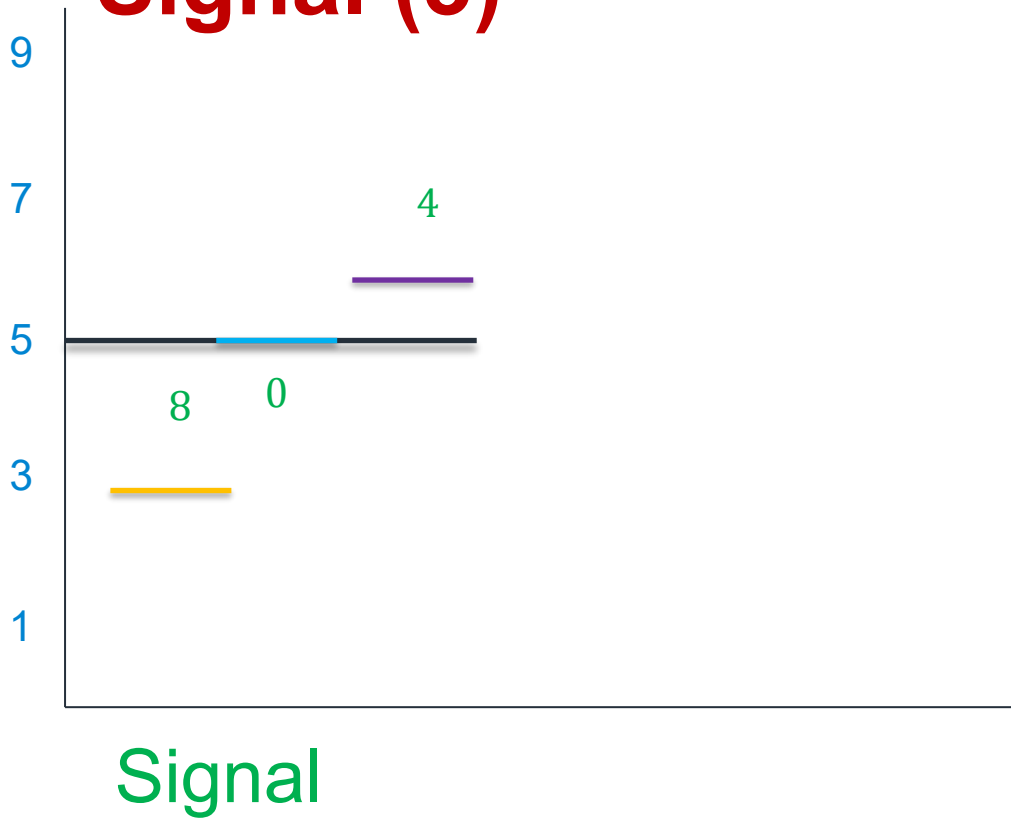
Signal (2)



We look at each distance, **squared**,

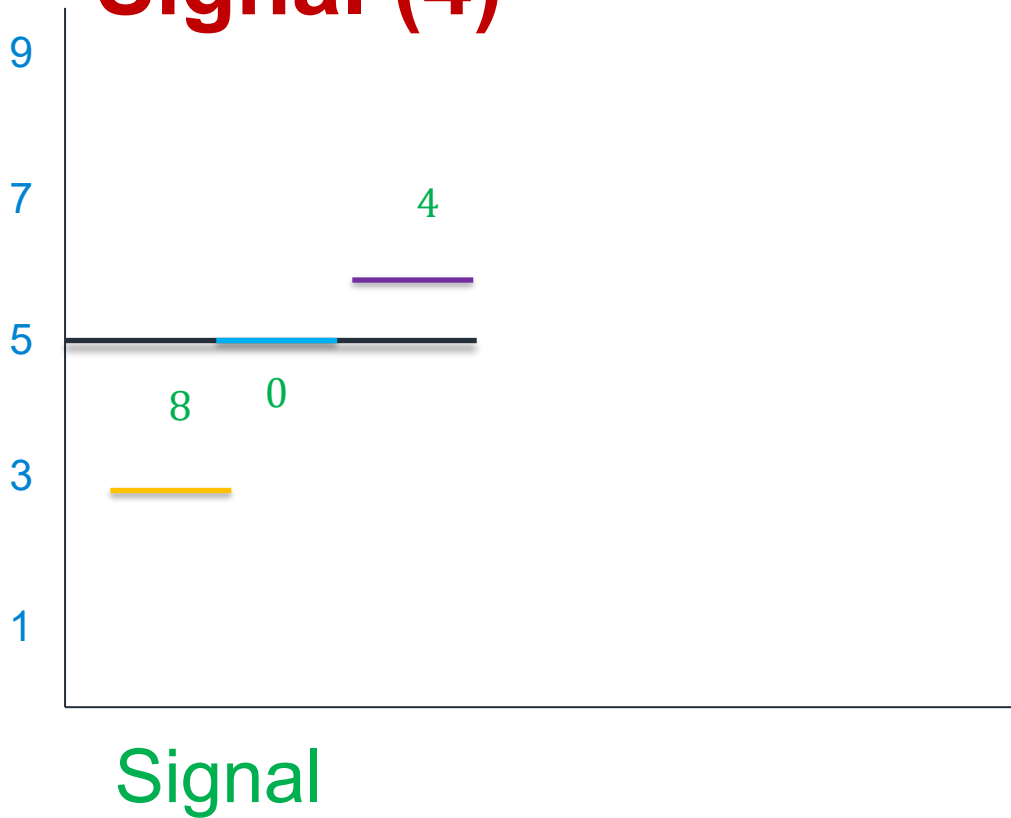
Signal

Signal (3)



We look at each distance, squared, “for each datapoint”, so that’s **twice**, **three times**, and **four times**.

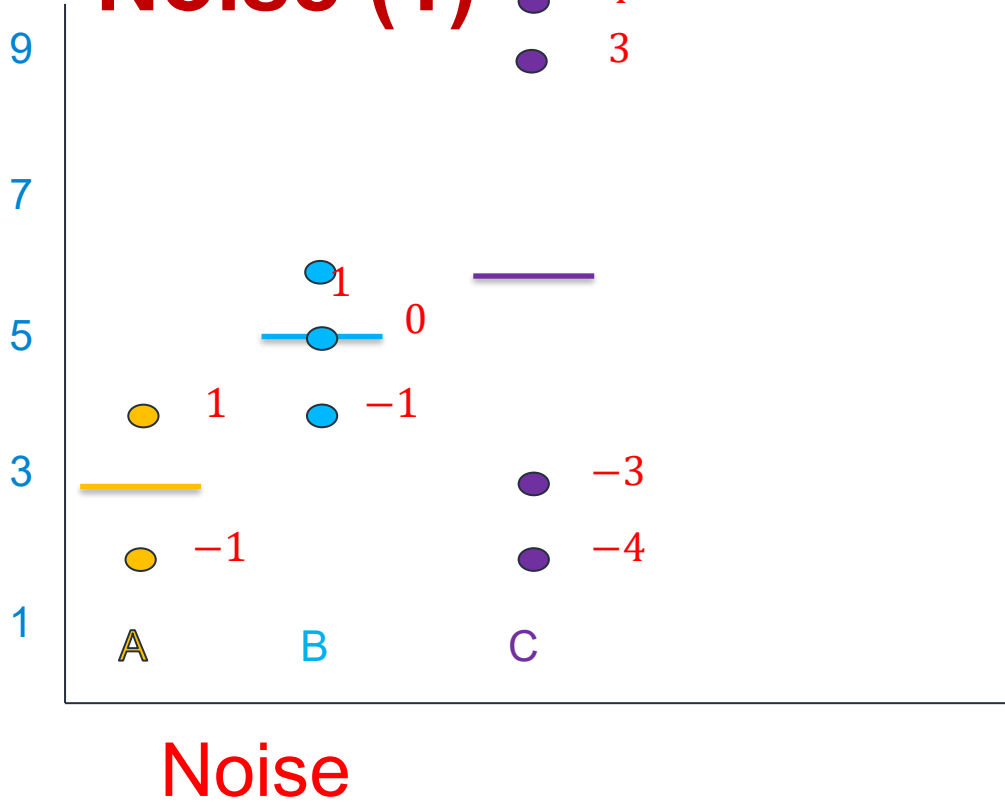
Signal (4)



We look at each distance, squared, “for each datapoint”, so that’s **twice**, **three times**, and **four times**.

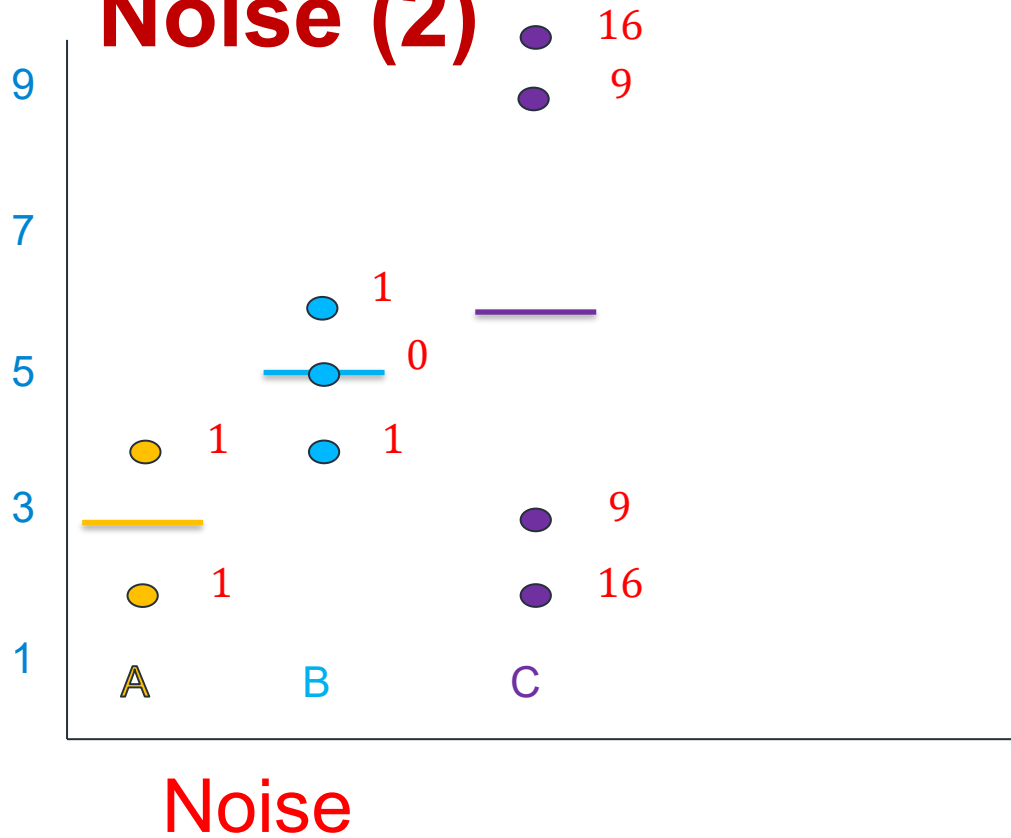
We sum them to get our **signal: 12**.

Noise (1)



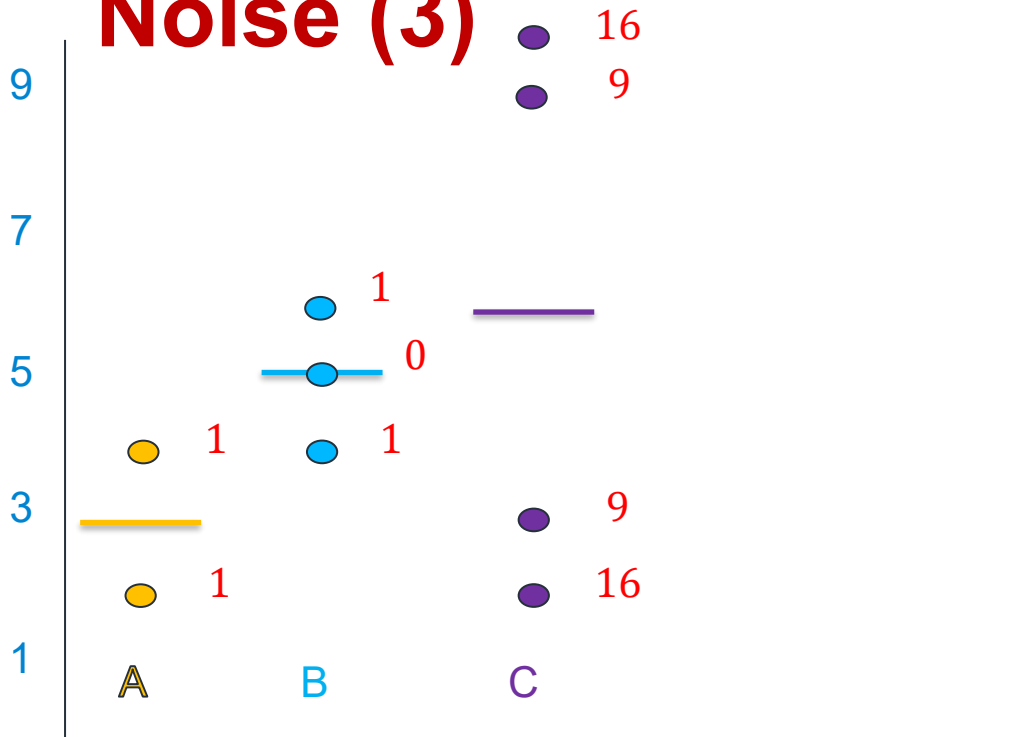
Again, we look at the distances between each datapoint and its group mean

Noise (2)



Again, we look at the distances, **squared**

Noise (3)



Noise

Again, we look at the distances, squared, and summed:

A: 2

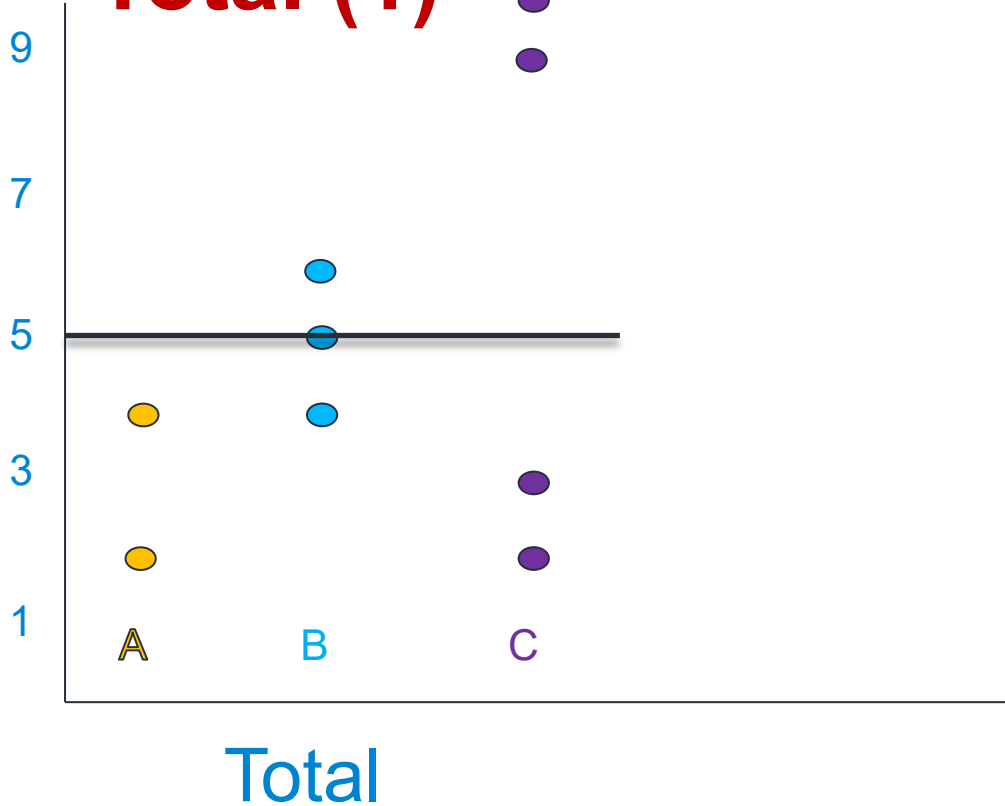
B: 2

C: 50

Noise: 54

(Quite a bit bigger than the signal...)

Total (1)

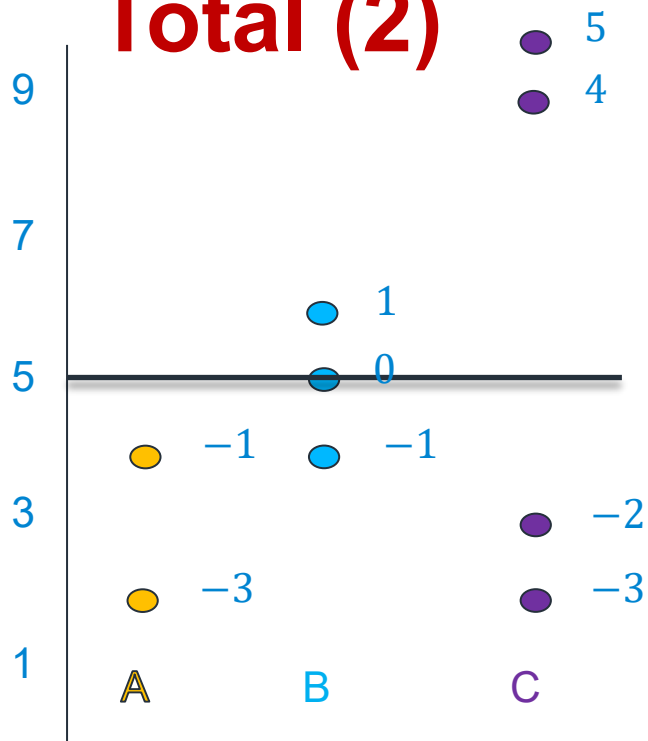


We don't technically need to do anything here, as

$$\text{Total} = \text{Signal} + \text{Noise}$$

but calculating the Total gives us a good check we have done things properly

Total (2)



Once again, it's the distances

Total

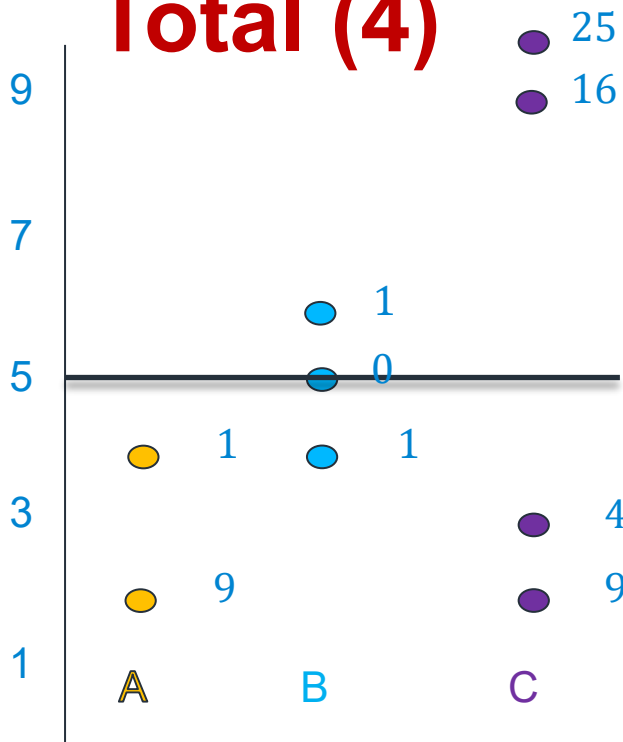
Total (3)



Once again, it's the
distances, squared

Total

Total (4)



Once again, it's the distances, squared, and summed:

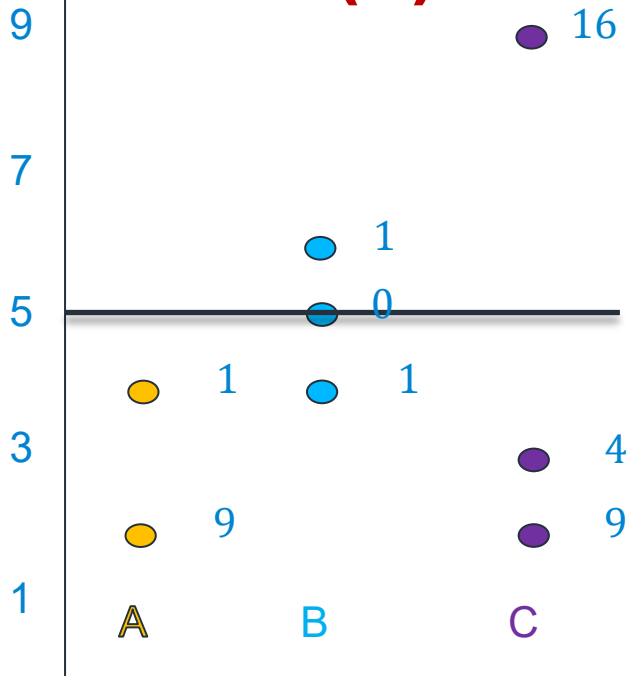
A: 10

B: 2

C: 54

Total: 66

Total (5)



Once again, it's the distances, squared, and summed:

$$A: 10 = 8 + 2$$

$$B: 2 = 0 + 2$$

$$C: 54 = 4 + 50$$

$$\text{Total: } 66 = 12 + 54$$

Total

Total = Signal + Noise

Total variation= (variation between group means) + (error variation not due to the group means)

$$SST = SSG + SSE$$

The next session: We finalise the test as:

Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

Source	SS	df	MSS	F
Groups (Signal)	12			
Errors (Noise)	54			
Total	66			

Source	SS	df	MSS	F
Groups (Signal)	12	2	6	0.667
Errors (Noise)	54	6	9	
Total	66	8		

From the previous lecture

$H_0: \mu_A = \mu_B = \mu_C$ **vs** $H_1: NOT H_0$

We had arrived here at the end of the last lecture:

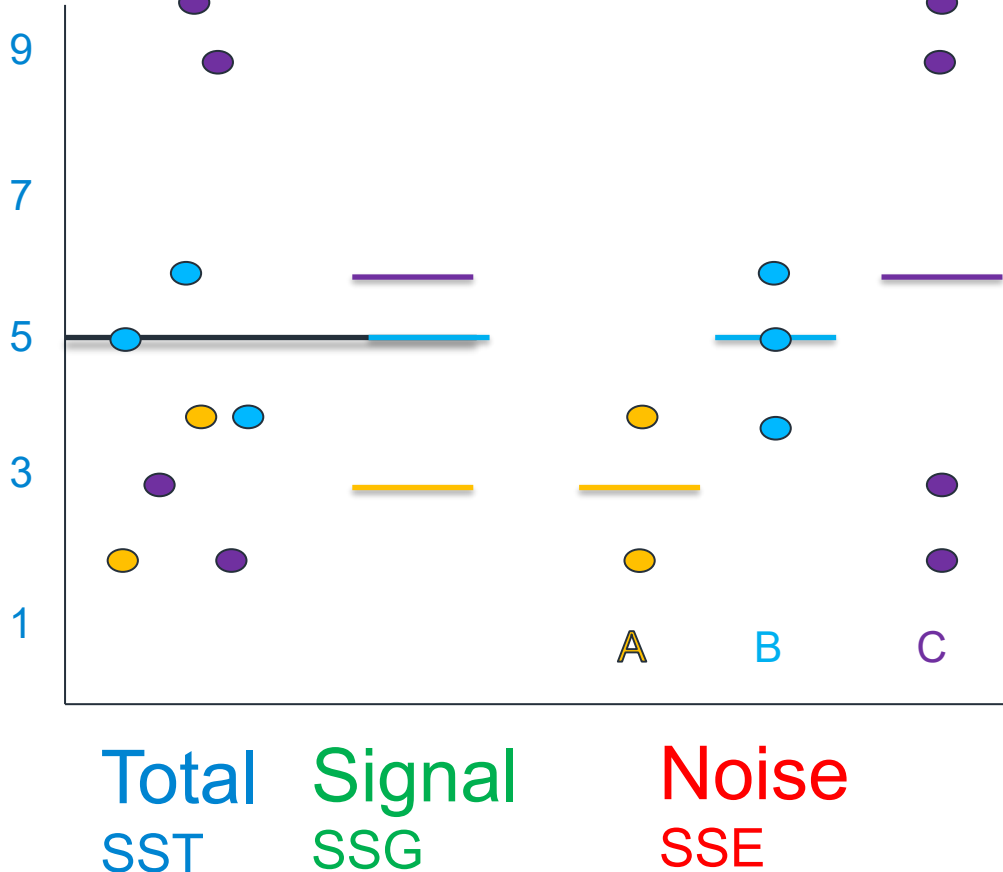
Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

Standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$



We had arrived here at the end of the last lecture:

Signal: sum of squared differences of each (observation's) group mean from the grand mean

Noise: sum of the squared differences of each datapoint from its group mean

Scores from formative tests (1)

Do students score differently across colleges?

Grand mean = $(2+4+2+5+6+2+3+9+10)/9=5$

Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

Group A

Alcuin College

Scores {2,4}

$$\bar{x} = (2+4)/2 = 3$$

$$\text{Signal} = 2(3-5)^2 = 8$$

$$\text{Noise} = (2-3)^2 + (4-3)^2$$

Group B

James College

Scores {4,5,6}

$$\bar{x} = (2+5+6)/2 = 5$$

$$\text{Signal} = ?$$

$$\text{Noise} = ?$$

Group C

Halifax College

Scores {2,3,9,10}

$$\bar{x} = 6$$

$$\text{SSG}_c = 2$$

$$\text{SSE}_c = 50$$

Scores from formative tests (2)

Do students score differently across colleges?

Grand mean= $(2+4+2+5+6+2+3+9+10)/9=5$

Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

Group A

Alcuin College

Scores {2,4}

$$\bar{x} = (2+4)/2 = 3$$

$$\text{Signal} = 2(3-5)^2 = 8$$

$$\text{Noise} = (2-3)^2 + (4-3)^2 = 2$$

Group B

James College

Scores {4,5,6}

$$\bar{x} = (4+5+6)/3 = 5$$

$$\text{Signal} = 3(5-5)^2 = 0$$

$$\text{Noise} = (4-5)^2 + (5-5)^2 + (6-5)^2 = 2$$

Group C

Halifax College

Scores {2,3,9,10}

$$\bar{x} = 6$$

$$\text{Signal} = 2$$

$$\text{Noise} = 50$$

Scores from formative tests (3)

Do students score differently across colleges?

Group A

Alcuin College

Scores {2,4}

$\bar{x} = 3$

Signal= 8

Noise=2

Signal=SSG

Noise=SSE

Total=SST

Group B

James College

Scores {4,5,6}

$\bar{x}=5$

Signal=0

Noise=2

Group C

Halifax College

Scores {2,3,9,10}

$\bar{x} = 6$

Signal =4

Noise=50

Source	SS	df	MSS	F
Groups (Signal)	12			
Errors (Noise)	54			
Total	66			

Degrees of Freedom (1)

Standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

In general, degrees of freedom (df) is the **number of values in the calculation of a statistic that are free to vary**.

In this module, it always arises from **calculation of s or SS**.

Type	T-statistic	Degrees of freedom
One-sample t-test	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	df = n - 1
Paired t-test	$t = \frac{\bar{X}_D - \mu_0}{s_D/\sqrt{n}}$	df = n - 1
Two-sample independent t-test:	$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$	df = n ₁ + n ₂ - 2

Degrees of Freedom (2)

Remember, we can manipulate the signal and the noise independently.

Source	SS	df	MSS	F
Groups (Signal)	12			
Errors (Noise)	54			
Total	66			

How many ways can I manage that?

Degrees of Freedom, Signal

I have 3 groups, so I can slide group A to wherever I want, so long as B and C counterbalance.

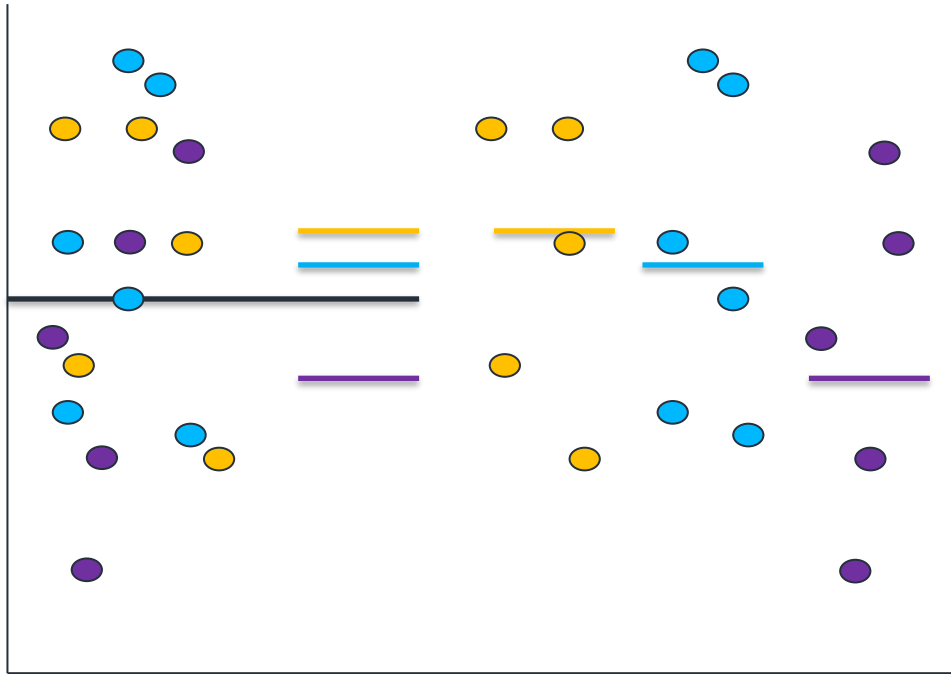
I can then slide B wherever I want, so long as C counterbalances.

A and B can both be set as I want without changing the noise.

$df(\text{Signal}) = \text{number of groups} - 1$

(Or I could set A and C, or B and C. Whichever, it's still two, one must counterbalance.)

From the previous session...



Total Signal Noise

$df(\text{Signal}) = df(\text{SSG}) =$
number of groups - 1

Source	SS	df	MSS	F
Groups (Signal)	12	2		
Errors (Noise)	54			
Total	66			

Degrees of Freedom, Noise

I have 2 data points in group A, so can set 1 where I want. 1 df.

... 3 in group B, so can set 2, so 2 df.

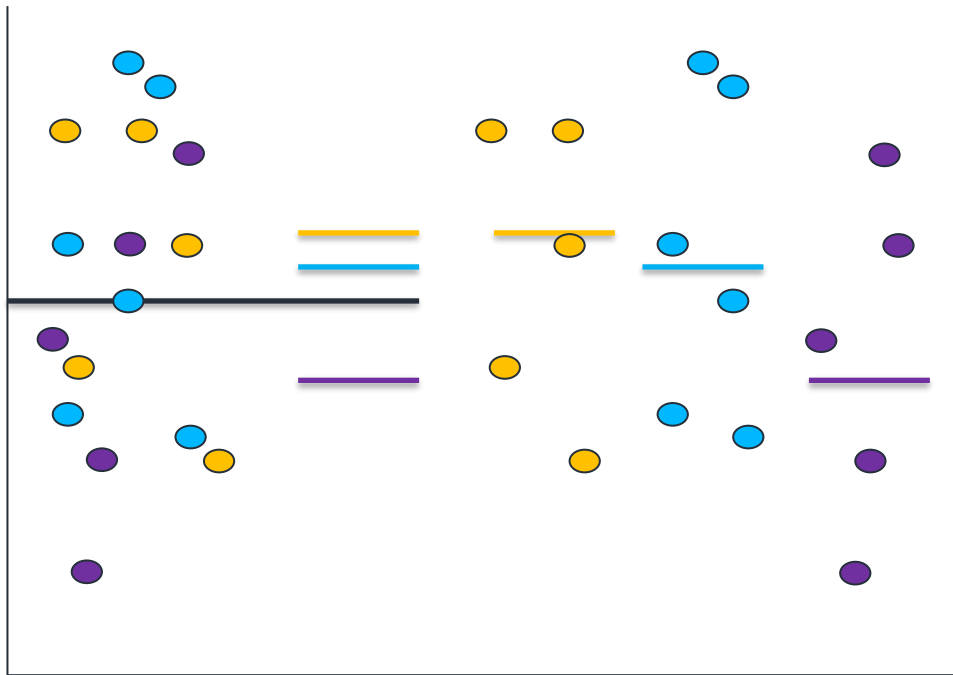
... 4 in C, so can set 3, so 3 df.

1 degree of freedom in the noise for A, 2 from B, 3 from C, so that's 6 in total.

$\text{df}(\text{Noise}) =$
number of datapoints – number of groups

We can add that to the table too.

From the previous session...



$df(\text{Noise}) = df(\text{SSE}) =$
number of datapoints – number
of groups

Total Signal Noise

Source	SS	df	MSS	F
Groups (Signal)	12	2		
Errors (Noise)	54	6		
Total	66			

Degrees of Freedom, Total

It seems reasonable then that the total degrees of freedom should be $2+6=8$.

And this is indeed the case.

In total, we have 9 datapoints (2 from A, 3 from B, 4 from C).

In exactly the same way, I can freely move 8 of them, so long as the 9th counterbalances.

8 freely chosen. 8 df.

Degrees of Freedom, summary

Source	SS	df	MSS	F
Groups (Signal)	12	2		
Errors (Noise)	54	6		
Total	66	8		

Degrees of Freedom, summary

We use the degrees of freedom to adjust our measures.

If there are three times as many ways to manipulate the **noise** as the **signal**, we might expect it to be three times as big?

We divide our measures (*the Sums of Squares (SS)*) by the degrees of freedom.

This gives the Mean Sums of Squares (MSS), a better measure of **signal** and **noise**.

Mean Sums of Squares

We have our better measures:

Source	SS	df	MSS	F
Groups (Signal)	12	2	6	
Errors (Noise)	54	6	9	
Total	66	8		

Lastly, F-test (1)

And then lastly we compare the **signal** to the **noise**.

How much bigger (or smaller) is the **signal** than the **noise**?

We do this through division.

$$MSS_{\text{signal}}/MSS_{\text{noise}}$$

Much as with the Coefficient of Variation, we care about the difference in a relative fashion, not an absolute one.

F-test (2)

Getting the Sums of Squares is the difficult part.

After that, the table continues.

And there we have the **Test Statistic!** It's called F-stat.

$$F\text{-stat} = \frac{SSG/df_G}{SSE/df_E}$$

Null distribution is $F(df_G, df_E)$

Source	SS	df	MSS	F
Groups (Signal)	12 +	2 +	= 6 =	0.667
Errors (Noise)	54 =	6 =	= 9 =	
Total	66	8		

F-test (3)

This gets compared to a **critical value**.

In this case $CV=5.14$. Since $TS < CV$, we **fail to reject the null**, and don't find a significant difference between the means.

(There isn't going to be a table distributed, the critical F-value will be given to you in this course.)

Group	Data				mean
A	2	4			3
B	4	5	6		5
C	2	3	9	10	6

Source	SS	df	MSS	F
Groups (Signal)	12	2	6	0.667
Errors (Noise)	54	6	9	
Total	66	8		

Steps for carrying out a hypothesis test

Step 1. Set up the null and alternative hypotheses

Step 2. Decide on the significance level (normally $p < 5\%$) and determine the critical values

Step 3. Calculate the test statistic

Step 4. Decide whether to accept or reject the null hypothesis

Step 1. $H_0: \mu_A = \mu_B = \mu_C$ vs $H_1: NOT H_0$

Step 2. 5%, CV (given) 5.14

Step 3. F-test from Anova

$F = 0.667$

Step 4. Unable to reject the hypotheses, $F < CV$, $0.667 < 5.14$

There is no significant difference between the means.

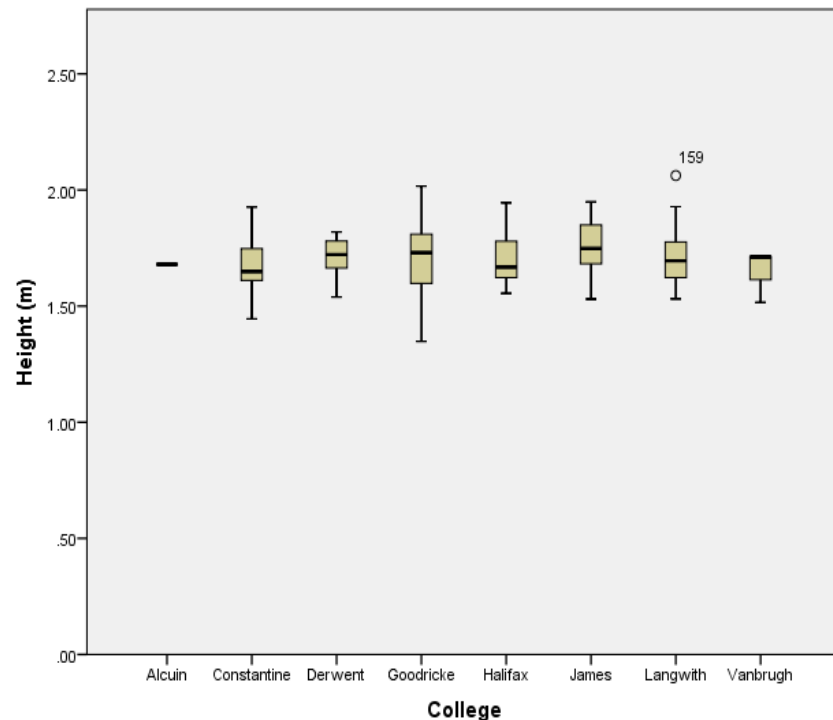
SPSS and ANOVA

We will do ANOVA in the computer practical sessions. Process will be faster but the output should be familiar!

ANOVA

Height (m)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.117	7	.017	1.040	.406
Within Groups	2.427	151	.016		
Total	2.544	158			



ANOVA

VAR00001

■

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	40.100	2	20.050	.279	.759
Within Groups	1580.540	22	71.843		
Total	1620.640	24			

Conclusions

- Step 1. Set up the null (no relationship, i.e. $\mu_A = \mu_C = \mu_D$) and alternative hypotheses
 - **If we reject the null, we find that at least one group is different to the rest.**
- Step 2. Decide on the significance level (normally $p < 5\%$) and determine the critical values
- Step 3. Calculate the test statistic (F-stat)
- Step 4. Decide whether to accept or reject the null hypothesis

Source	SS	df	MSS	F
Groups (Signal)				
Errors (Noise)				
Total				

Next session: Another example and Anova using calculator

Short cut – using your calculator

Instead of the point by point calculation of the distance, squaring, and then summing,

we can make use of the definition of **sample variance** s^2 :

$$\sum (x - \bar{x})^2 = (n - 1)s^2$$

1 : $\bar{x}$	2 : σ^2_x
3 : σ_x	4 : s^2_x
5 : s_x	6 : n

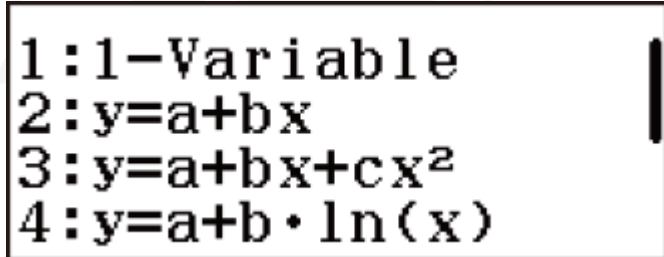
$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sigma_n^2 = \frac{\sum (x - \bar{x})^2}{n}$$

Short cut – using your calculator

Perform the steps below to start a statistical calculation.

1. Press **MENU**, select the Statistics Mode icon, and then press **=**.



1:1-Variable
2:y=a+bx
3:y=a+bx+cx²
4:y=a+b•ln(x)

2. Press **1**, select 1-Variable



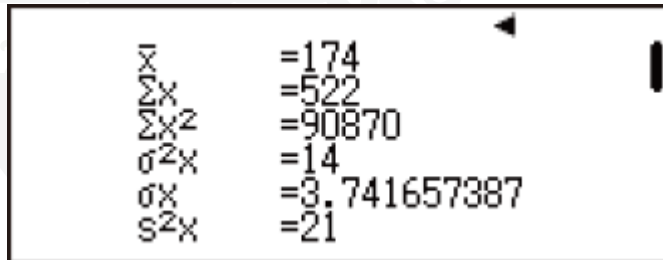
	x	Freq
1		
2		
3		
4		

Short cut – using your calculator

3. Input a value and press .

Repeat this step until all the datapoints are inputted.

4. Display statistical values based on input data. Press 
Then select **1-Variable Calc**



$\bar{x}$	=174
Σx	=522
Σx^2	=90870
$\sigma^2 x$	=14
σx	=3.741657387
$s^2 x$	=21

[CASIO Statistical Calculations \(85GT/X\)](#)

Short cut – using your calculator

It is more convenient to use the estimate for population variance σ^2

$$\sum (x - \bar{x})^2 = n\sigma_n^2$$

1: $\bar{x}$	2: $\sigma^2 x$
3: σx	4: $s^2 x$
5: $s x$	6: n

$\bar{x}$	=174
$\sum x$	=522
$\sum x^2$	=90870
$\sigma^2 x$	=14
σx	=3.741657387
$s^2 x$	=21

Note, we're not using σ and s as standard deviations, just as shortcuts to calculate $\sum (x - \bar{x})^2$. Therefore it doesn't matter which you use. **Common to use $n\sigma^2$ because it's slightly faster.**

Applying this to our example

So, how do we actually use this concept?

Recall the data are:

Group	Data			
A	2	4		
B	4	5	6	
C	2	3	9	10

We just put in different combinations of these (and the group means), calculate $n\sigma_n^2$ for each combination, and ...

Applying this to our example

Group	Data			
A	2	4		
B	4	5	6	
C	2	3	9	10

First up, put all 9 in.

We will want $\bar{x}$ and $n\sigma_n^2$ (this is for the Total).

You should get **5** and **66**.

Applying this to our example

Group	Data			
A	2	4		
B	4	5	6	
C	2	3	9	10

First up, put all 9 in.

We will want $\bar{x}$ and $n\sigma_n^2$
(this is for the Total SST).

You should get 5 and 66.

Then, we put each group in individually,
and again get

$$\bar{x} \text{ and } n\sigma_n^2.$$

You should find

A gives 3 and 2,

B gives 5 and 2, and

C gives 6 and 50.

$2 + 2 + 50 = 54$, which is the noise.

Applying this to our example

Group	Data			
A	2	4		
B	4	5	6	
C	2	3	9	10

Lastly, we put in the means with the group sizes as frequencies:

Mean	Freq
3	2
5	3
6	4

Again, find $\bar{x}$ and $n\sigma_n^2$.

You should find 5 and 12.

Note that this mean equals the mean from all 9 datapoints.

Also, $66 = 54 + 12$ (again).

Applying this to our example

This all puts us in the same place as after the diagrams:

Finding the degrees of freedom, mean sums of squares and the test statistic remain exactly the same.

Source	SS	df	MSS	F
Groups (Signal)	12			
Errors (Noise)	54			
Total	66			

Last time – summary

Source	SS	df	MSS	F
Groups (SSG)	12 +	2 +	= 6 =	0.667
Errors (SSE)	54 =	6 =	= 9	
Total (SST)	66	8		

Exercise – the training times needed for the three systems (1)

An important factor in selecting software for word processing and database management systems is the time required to learn how to use a particular system. In order to evaluate three file management systems, a firm devised a test to see how many training hours were needed for five of its word processing operators to become proficient in each of three systems.

System A	16	19	14	13	18
System B	16	17	13	12	17
System C	24	22	19	18	22

Using a 5% level, is there any difference between the training times needed for the three systems? (CV=3.89)

Exercise – the training times needed for the three systems (2)

- **Total sum of squares = Signal + Noise**
- In practice we calculate **Total SS** and **Signal** and find **Noise** by difference:
Noise = Total SS - Signal

$$s_n = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

- The classical method of finding these sums of squares is very laborious so, instead, we shall make use of the fact that we can produce a standard deviation s_n from a calculator.

so the 'total sum of squares', $SST = \sum (x - \bar{x})^2 = (n - 1)s_n^2$

- from calculator using $[s^2x]$ to give us s_n^2 . ($[\sigma^2x]$ to give us σ_n^2)
-)

Exercise – the training times needed for the three systems (3)

Total SS: Input all the data values individually and output the values for n , and s_n (σ_n) from the calculator in **SD mode**. Use their values to calculate σ_n^2 and $n\sigma_n^2$.

n	$\bar{x}$	σ_n	σ_n^2	$n \sigma_n^2$
15	17.33	.419	11.69	<u>175.3 = SST</u>

(If s_n is output, its square is multiplied by $(n-1)$.)

Signal Calculate n and $\bar{x}$ for each of the systems separately:

	n	$\bar{x}$
▪ System A	5	16
▪ System B	5	15
▪ System C	5	21

Exercise – the training times needed for the three systems (4)

- Use these values as **frequency data** and output the values for n , $\bar{x}$ and s_n (or σ_n) as before. Use these values to calculate $n s_n^2$

- | | | | | | |
|------------------|-----|-----------|------------|--------------|----------------|
| | n | $\bar{x}$ | σ_n | σ_n^2 | $n \sigma_n^2$ |
| SS Signal | 15 | 17.33 | 2.625 | 6.889 | <u>103.3</u> |
- SS Noise** = Total - Signal = $175.3 - 103.3 = \underline{72.0}$

- ANOVA Table:**

Source	S.S.	d.f.	MSS	F
Between systems	103.3	$3 - 1 = 2$	$103.3/2$ $= 51.65$	$51.65/6.00$ $= 8.61$
Errors	72.0	$14 - 2 = 12$	$72.0/12$ $= 6.00$	
Total	175.3	$15 - 1 = 14$		

Exercise – Solution

$$H_0: \mu_A = \mu_B = \mu_C$$

H_1 : At least two means are different.

Critical value: $F_{0.05}(2,12) = 3.89$

Test Statistic: 8.60

Conclusion: T.S. > C.V. so reject H_0 . There is a difference between at least two of the systems.

Conclusions from ANOVA

ANOVA tests if there is a significant difference between the means of the groups.

We can either:

- find a difference
- not find a difference

- If we do, we still don't know which group(s) was/were different, nor whether it/they was/were more or less than the others.

Tests do exist to work this out. ANOVA typically covered again in more advanced stats courses.